

YHLO

LIS Protocol Manual

Version V1.1

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1. OVERVIEW

The LIS interface described in this document is used for data exchange between the testing instrument and laboratory information system (LIS).Through this interface, the instrument can download the sample application information from the LIS server through a variety of query methods, or automatically and real-time, or send the sample results individually and in batches manually, and the quality control results to the LIS server.LIS server can actively send the application information of samples to the instrument.

For the convenience of description, in the following paper, the instrument refers to the inspection equipment of aifl and LIS refers to the hospital laboratory information system.

1.1. Scope of application

Complete LIS functions supported include:

- support real-time results to upload automatically in the process of testing sample LIS server.
- support user through history query result, a single or batch will upload the LIS server sample results.
- support users to control the quality by query results, will upload the LIS server quality control results
- support through the sample bar code from the LIS server query sample information;
- support through the sample number from the LIS server query sample information;
- support by specifying the time period, from the LIS server query sample information;
- support LIS active distributed application server information;

Function	iFlash Series	Other Product
Real-time sample results upload	support	/
Upload historical sample results	support	/
Quality control results uploaded	support	/
Query from LIS by bar code	support	/
Query from LIS by sample number	support	/
Query from LIS by time period	HL7 Protocol support	/
LIS server test application on its own initiative	support	/

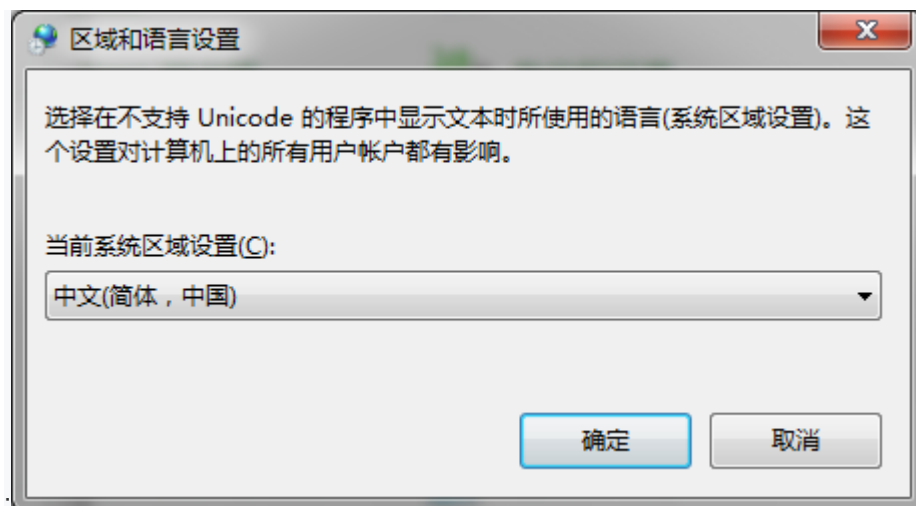
iFlash series models include: iFlash3000 A/B/C/H/I

1.2. LIS physical connection

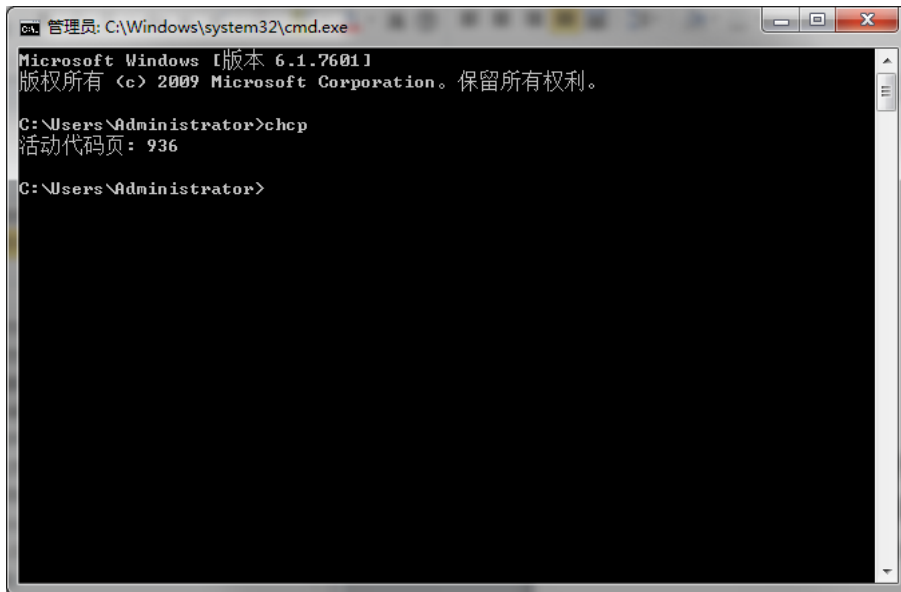
The protocol itself supports both RS232 and TCP/IP physical connections. The hardware environment that needs software to run at the same time is configured to have serial port, network port.

1.3. Supported code encoding

Support UTF8, ASCII encoding, can be configured in software. Among them, ASCII encoding refers to the default character encoding of the operating system, which depends on the region and language Settings in the operating system of the instrument end. Typical cases, such as simplified Chinese WIN7 system, are set as follows (corresponding to GBK code)



Code page: 936



2.HL7 Protocol

2.1. OverviewProtocol version

This Interface is defined based on the HL7 v2.3.1 protocol. Please refer to HL7 Interface Standards Version 2.3.1 for the full contents of the standard.

2.1.2. Supported LIS messages

The HL7 messages used by this interface are only ORU, ACK, QRY, QCK, DSR.

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2.1.3. Basic format of HL7 protocol

According to the Minimal Lower Layer Protocol (MLLP) and the description of HL7 Interface Standards Version 2.3.1,

The HL7 message is transmitted in the following format:

<SB> ddddd <EB><CR>

➤ <SB>

<SB>=Start Block character (1 byte)

One byte, is hexadecimal characters <0x0B>, ASCII<VT>。

➤ **dddd**

dddd = Data (variable number of bytes)

dddd is HL7 message content. Only includes ISO 8859-1 characters (Hexadecimal value 20 - FF) and <CR>, not include other controls and characters that cannot be printed。

➤ **<EB>**

<EB>= End Block character (1 byte)

One byte, is hexadecimal characters <0x1C >, ASCII <FS>。

➤ **<CR> = Carriage Return (1 byte)**

<CR> = Carriage Return (1 byte)

One byte, is hexadecimal characters <0x0D >, ASCII <CR>。

2.1.4. HL7 message syntax

Each HL7 message consists of segments that end with <CR>.

Each segment consists of a three fields and fixed number of domains, The domain consists of components and subcomponents, Define the delimiter of each component in the MSH segment of each message.

For example:

MSH|^~\&|YHLO|iFlash3000|||20160101235959||ORU^R01|1|P|2.3.1|||0||ASCII||

The five character after MSH is used to distinguish the separators of domains, components, and subcomponents. YHLO HL7 standard uses the characters of the following table:

Character	Meaning
	Field separator
^	Component separator
&	Subcomponent separator

~	Duplicate separator
\	Escape character

The first domain of MSH includes all separators. Some of the latter domains are blank because they are optional. Currently YHLO HL7 interface does not use it, so it is blank.

Domain 9:	Contains message types and events (ORU、R01)
Domain 10:	Contains a message that uniquely identifies the message ID
Domain 11:	Contains the processing ID (P indicates product)
Domain 12:	Define the HL7 version of the message used (2.3.1)

For any message, the order after the MSH segment is specified, These sequences are described in detail in the following sections, using these syntax constructs to indicate that segments are optional or repeatable:

[] Indicates that the segment inside is optional.

{ } Indicates that each segment can be repeated 0 times or 1 or more times.

2.1.5. Character transfer

: In string type field data, such as remarks, diagnostic information, user-defined gender and other string data, escape delimiters may appear. The delimiters in the original string should be escaped into an escape character sequence when encoding, and then restored when decoding. HL7 interface USES escape rules as follows:

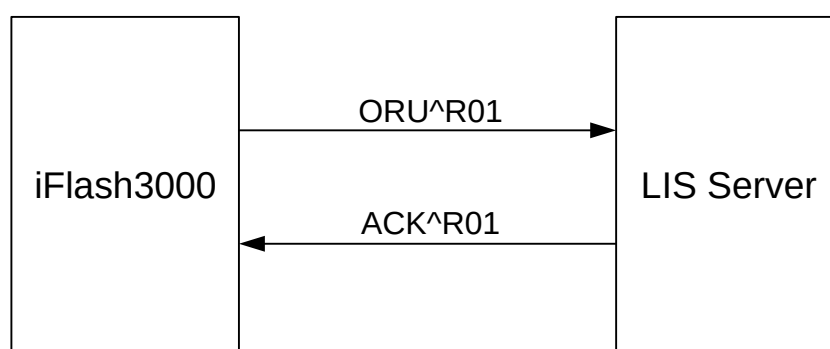
Escape character sequences	The original character
\F\	Field separator ' '
\S\	The component separator '^'
\T\	Subcomponent separator '&'
\R\	Repeat separator '~'
\E\	Escape separator '\'

\\.br\\	Message segment terminator <CR>
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2.2. Conversation process

2.2.1. Test results uploaded to the LIS server

2.2.1.1.



Abbreviation:

ORU: Observational Results (Unsolicited)

ACK: Acknowledgment

Explanation:

ORU: For instrument uploading test results, It triggers event R01.

ACK: For LIS server response to ORU messages.

2.2.1.2. ORU

Patient sample (including quality control sample) test result / response

The instrument uses the ORU message to send the test result to the LIS server, LIS server must reply to the instrument with an ACK message after successfully receiving the test results, otherwise the instrument will think that LIS did not receive the test results successfully.

2.2.1.3. Sample test result

For the sample test results, the specific structure of the ORU message is as follows:

ORU	Description
MSH	Message Header
PID	Patient information (patient name, sample number, etc.)
OBR	Observation report (sample type, sending doctor, laboratory doctor, clinical diagnosis, etc.)
{OBX}	Test result

If there are more than one item in a sample, the message length will be increased accordingly. In order to improve the communication reliability, if there are more than 50 items in a sample, the instrument will automatically send messages in groups.

2.2.1.4. QC test result

For the QC test results, the specific structure of the ORU message is as follows:

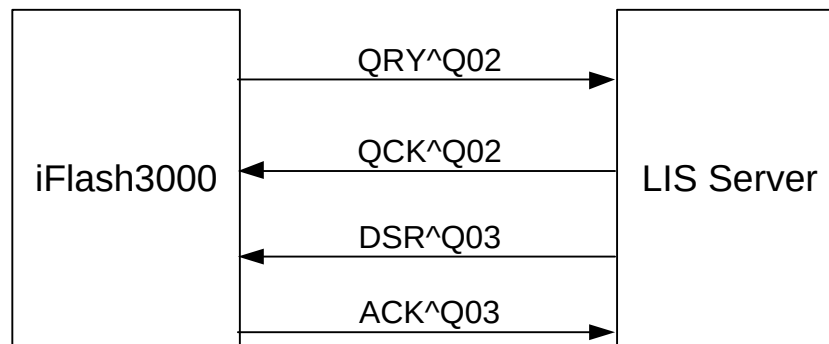
ORU	Description
MSH	Message Header
OBR	1. Item information (item number, item name) 2. Observation report (quality control information: number, number, name, batch number, expiration date, average) 3. QC time, test result (concentration value)

For quality control results, a message will send all the quality control test results of a item.

2.2.1.5. ACK

ACK	Description
MSH	Message Header
MSA	Message Acknowledgment Segment

2.2.2. Get sample information from the LIS server



Abbreviation:

QRY: Query

QCK: Query Acknowledgment

DSR: Display Response

ACK: Acknowledgment

Explanation:

QRY is used to inquire the application information to the LIS system, which triggers event Q02.

QCK message is used to respond to the QRY message.

The main function of the DSR message is to let the LIS server send sample request information to the instrument.

The ACK message is used to respond to the DSR message.

2.2.2.1. QRY

QRY message structure is as follows:

QRY	Description
MSH	Message Header

QRD	Query Definition Segment
QRF	Query Filter Segment

2.2.2.2. QCK

QCK message structure is as follows:

QCK	Description
MSH	Message Header
MSA	Message Acknowledgment Segment
ERR	Error Segment
QAK	Query Acknowledgment Segment

2.2.2.3. DSR

DSR message structure is as follows:

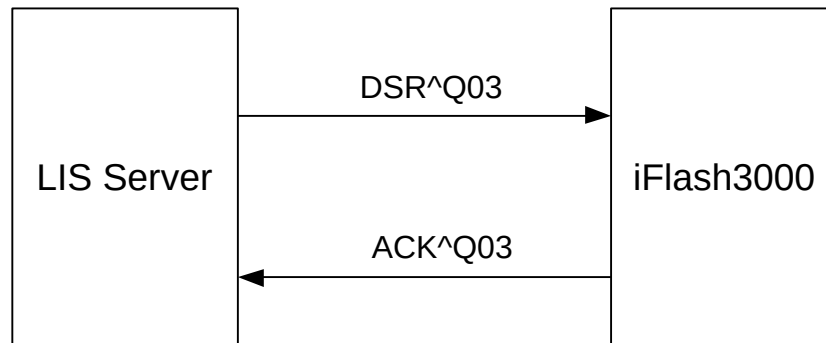
DSR	Description
MSH	Message Header
MSA	Message Acknowledgment Segment
ERR	Error Segment
QAK	Query Acknowledgment Segment
QRD	Query Definition Segment
QRF	Query Filter Segment
{DSP}	Display Data Segment
DSC	Continuation Pointer Segment

2.2.2.4. ACK

ACK message structure is as follows:

DSR	Description
MSH	Message Header
MSA	Message Acknowledgment Segment
ERR	Error Segment

2.2.3 LIS server sends sample application information automatically



Abbreviation:

DSR: Display Response

ACK: Acknowledgment

Explanatio: See 2.2.3 description of DSR, 2.2.4 description ACK

2.3 Message segment

The following is a detailed description of each message segment, where the serial number has the mandatory field specified by HL7, the remaining fields are optional, and the interface is a subsequent extension. If the optional field is not needed, it is not omitted, but is blank.

2.3.1 MSH - Message Header

The first message segment of HL7 message, all messages start with MSH segment, and are usually located at the front of the message segment.

This message segment is used to define the intention, source, purpose, and some details of the message syntax.

The MSH segment of the YHLO HL7 interface uses the following domains:

Serial number	Field name	Length	Exlanation
1#	Field Separator	1	The separator between the segment ID and the first real domain, which defines the domain separator for the rest of the message ().
2#	Encoding	4	Contains component delimiter, repeating

Serial number	Field name	Length	Exlanation
	Characters		delimiter, escape separator, and child component separator. (^~\&)
3	Sending Application	180	Sender application, use YHLO.
4	Sending Facility	180	Sender device,use device model, like iFlash 3000.
5	Receiving Application	180	Blank,Retain.Receiver application.
6	Receiving Facility	180	Blank,Retain.Receiver device.
7	Date/Time Of Message	26	The time of the current message. Call system time information.
8	Security	40	Blank, Retain. Security.
9#	Message Type	7	Message Type,like:ORU^R01
10#	Message Control ID	20	Message Control ID , Uniquely identifies a message, increasing with the number of messages from 1.
11#	Processing ID	3	Processing ID,used “P” (Represent the product)
12#	Version ID	60	Version ID, HL7 protocol version: V2.3.1
13	Sequence Number	15	Blank, Retain.
14	Continuation Pointer	180	Blank, Retain.
15	Accept Acknowledgment Type	2	Receive response type, valid only for DSR^Q03 messages. The value is P, which indicates the sample information sent by the LIS system. Other values indicate sample information in response to the LIS query, can be blank.
16	Application Acknowledgment Type	2	As the result type of the transmission.0- 病 Patient sample test results;1- Calibration result;2-QC result.
17	Country Code	2	Blank, Retain.
18	Character Set	10	ASCII

Serial number	Field name	Length	Exlanation
19	Principal Language Of Message	60	Blank, Retain.
20	Alternate Character Set Handling Scheme	20	Blank, Retain.

Remarks: This segment will appear in all messages. In the HL7 message sent by the LIS system to the instrument, the values of fields 3 and 4 of the paragraph are determined by the LIS developer, and the values of fields 5 and 6 are "YHLO" and the device model respectively. The tenth, sixteenth field uses integer type; all other fields use string type. When the LIS system initiatively sends the sample application information to the instrument, the fifteenth field needs to take the value of "P".

2.3.2 MSA - Message Acknowledgment Segment

The MSA segment of the YHLO HL7 interface uses the following domains:

Serial number	Field name	Length	Exlanation
1#	Acknowledgment Code	2	AA-Accepts; AE-Error; AR-Refuse. It should be noted that if the LIS sends a response test message, that is, ACK^R01 message, if the LIS uses AE or AR, the instrument will consider that the LIS has not processed the result of the transmission, and will re-send the test result, the maximum attempt is 2 times. So it is recommended that LIS not use AE and AR, only use AA.
2#	Message Control ID	20	Same as the MSH message control ID and the same as the sender's MSH-10.
3	Text Message	80	A textual description of the event. Corresponds to the sixth

			field. Can be used to write error logs.
4	Expected Sequence Number	15	Blank, Retain.
5	Delayed Acknowledgment Type	1	Blank, Retain.
6	Error Condition	100	Status code

Explanation: The values of the MSA-6 field are shown in the table below.

Status code (MSA-6)	Status text (MSA-3)	Description / Notes
Success Status code:		AA
0	Message accepted	Success
Error Status code:		AE
100	Segment sequence error	The order of the message segments is incorrect, or the required segments are missing.
101	Required field missing	Missing required fields in one segment
102	Data type error	The data type of the field is incorrect, for example, the number becomes a character
103	Table value not found	Indicates that the value was not found, temporarily not used
Refuse Status code:		AR
200	Unsupported message type	Message type is not supported.
201	Unsupported event code	Event code is not supported.
202	Unsupported	Processing ID is not supported

Status code (MSA-6)	Status text (MSA-3)	Description / Notes
	processing id	
203	Unsupported version id	Version ID is not supported
204	Unknown key identifier	Unknown keyword identification, such as transmitting a patient information that does not exist
205	Duplicate key identifier	Duplicate keywords already exist
206	Application record locked	The transaction cannot be executed at the application storage level, as the database is locked
207	Application internal error	Unknown application internal error

Remarks:The message segment can appear in ACK^R01, QCK^Q02 and ACK^Q03 messages. The fourth, sixth field uses integer type; other fields use string type.

2.3.3 PID – Patient Identification

PID segment is mainly used to build personal information of patients. The PID segment of the YHLO HL7 interface uses the following domains:

Serial number	Field name	Length	Exlanation
1	Set ID – PID	10	Identify message segments for different patients
2	Patient ID	20	admission number
3#	Patient Identifier List	20	Medical record No
4	Alternate Patient ID – PID	20	Hospital bed number
5#	Patient Name	48	Patient Name
6	Mother's	48	Hospital ward

Serial number	Field name	Length	Exlanation
	Maiden Name		
7	Date/Time of Birth	26	Patient date of birth (Format: year, month, day, hour, minute, and second. Such as 19830512000000) ,can be blank.
8	Sex	1	Male, send M Female, send F Other, send O Unknown, sned U
9	Patient Alias	48	Blood type (A-'A', B-'B', AB-'AB', O-'O',)
10	Race	80	Blank, Retain.
11	Patient Address	106	Patient Address
12	County Code	4	County Code (Zip code)
13	Phone Number - Home	40	Phone Number - Home
14	Phone Number - Business	40	Blank, Retain.
15	Primary Language	60	Blank, Retain.
16	Marital Status	80	Blank, Retain.
17	Religion	80	Blank, Retain.
18	Patient Account Number	20	/
19	SSN Number -Patient	16	/
20	Driver's License Number –	25	/

Serial number	Field name	Length	Exlanation
	Patient		
21	Mother's Identifier	20	Blank, Retain.
22	Ethnic Group	80	/
23	Birth Place	60	/
24	Multiple Birth Indicator	1	Blank,Retain.Multi-child indicator, yes-Y, no-N
25	Birth Order	2	Blank, Retain.Birth order, an integer greater than 0
26	Citizenship	80	Remarks
27	Veterans Military Status	60	Blank, Retain.
28	Nationality	80	/
29	Patient Death Date and Time	26	Blank, Retain.
30	Patient Death Indicator	1	Blank, Retain. Patient death indicator, yes-Y, no-N

Remarks:This message segment is used only in ORU^R01 messages. Fields 1 and 25 use integer data types; fields 24 and 30 use Boolean data types; other fields use string data types。

2.3.4 OBR – Observation Request

OBR used for transfer some information about report from doctor.

2.3.4.1 Patient sample test results

When transfer patient sample test results information（MSH-16 is 0 时），bellows are some parameters field under YHLO HL7 interface.

NO.	field	length	Description
1	Set ID – OBR	10	Confirm different OBR field

NO.	field	length	Description
2	Placer Order Number	22	Used for sample barcode number
3	Filler Order Number	22	Used for sample serial number
4#	Universal Service ID	200	Universal Service ID
5	Priority	2	Whether is emergency, yes is Y, or N
6	Requested Date/time	26	Sample collection time
7	Observation Date/Time	26	Observation Date/Time
8	Observation Date/Time End	26	Observation End Date/Time
9	Collection Volume	20	Used for manual dilution of the sample
10	Collector Identifier	60	Used for sample position ID (combined by rack number and position number)
11	Specimen Action Code	1	Specimen Action Code
12	Danger Code	60	Danger Code
13	Relevant Clinical Info.	300	Used as clinical diagnostic information for patients
14	Specimen Date/Time Received	26	Specimen Received Date/Time
15	Specimen Source	300	different sample types "serum" , "plasma" and "urine"
16	Ordering Provider	120	Ordering Provider
17	Order Callback Phone Number	40	Order Callback Phone Number
18	Placer Field 1	60	sample properties include "icterus", "hemolysis", and "lipemia".
19	Placer Field 2	60	Blood bag number
20	Filler Field 1	60	Attending doctor

NO.	field	length	Description
21	Filler Field 2	60	Treatment of department
22	Result Rpt/Status Change – Date/Time	26	Result Rpt/Status Change – Date/Time
23	Charge to Practice	40	Charge to Practice
24	Diagnostic Serv Sect ID	10	Diagnostic Serv Sect ID
25	Result Status	1	Result Status
26	Parent Result	200	Parent Result
27	Quantity/Timing	200	Quantity/Timing
28	Result Copies To	150	Result Copies To
29	Parent	150	Parent
30	Transportation Mode	20	Transportation Mode
31	Reason for Study	300	Reason for Study
32	Principal Result Interpreter	200	Principal Result Interpreter
33	Assistant Result Interpreter	200	Assistant Result Interpreter
34	Technician	200	Technician
35	Transcriptionist	200	Transcriptionist
36	Scheduled Date/Time	26	Scheduled Date/Time
37	Number of Sample Containers	4	Number of Sample Containers
38	Transport Logistics of Collected Sample	60	Transport Logistics of Collected Sample
39	Collector's Comment	200	Collector's Comment
40	Transport Arrangement Responsibility	60	Transport Arrangement Responsibility
41	Transport Arranged	30	Transport Arranged
42	Escort Required	1	Escort Required
43	Planned Patient Transport Comment	200	Planned Patient Transport Comment
44	Ordering Facility Name	60	Ordering Facility Name

NO.	field	length	Description
45	Ordering Facility Address	106	Ordering Facility Address
46	Ordering Facility Phone Number	48	Ordering Facility Phone Number
47	Ordering Provider Address	106	Ordering Provider Address

Note: The news section is only used to ORU ^ R01 message. Fields 1, 3, and 37 use integer data types; Field 9 USES floating point data types; All other fields use string data types.

2.3.4.2 Quality control test results

If the results of the project quality control test are transmitted (msh-16 is 2).Its fields are defined as follows:

No.	field	length	Description
1	Set ID – OBR	10	Confirm different OBR field
2	Placer Order Number	22	Placer Order Number
3	Filler Order Number	22	Filler Order Number
4#	Universal Service ID	200	Universal Service ID
5	Priority	2	Priority
6	Requested Date/time	26	Requested Date/time
7	Observation Date/Time	26	Observation Date/Time
8	Observation End Date/Time	26	Observation End Date/Time
9	Collection Volume	20	Collection Volume
10	Collector Identifier	60	Collector Identifier
11	Specimen Action Code	1	QC quantity
12	Danger Code	60	QC serial number
13	Relevant Clinical	300	QC name

No.	field	length	Description
	Info.		
14	Specimen Received Date/Time	26	QC lot number
15	Specimen Source	300	QC warranty period
16	Ordering Provider	120	Ordering Provider
17	Order Callback Phone Number	40	QC concentration level。 H; M; L
18	Placer Field 1	60	QC mean value
19	Placer Field 2	60	QC standard deviation
20	Filler Field 1	60	Test result value
21	Filler Field 2	60	Empty
22	Result Rpt/Status Change – Date/Time	26	Result Rpt/Status Change – Date/Time
23	Charge to Practice	40	Charge to Practice
24	Diagnostic Serv Sect ID	10	Diagnostic Serv Sect ID
25	Result Status	1	Result Status
26	Parent Result	200	Parent Result
27	Quantity/Timing	200	Quantity/Timing
28	Result Copies To	150	Result Copies To
29	Parent	150	Parent
30	Transportation Mode	20	Transportation Mode
31	Reason for Study	300	Reason for Study
32	Principal Result Interpreter	200	Principal Result Interpreter
33	Assistant Result Interpreter	200	Assistant Result Interpreter
34	Technician	200	Technician
35	Transcriptionist	200	Transcriptionist

No.	field	length	Description
36	Scheduled Date/Time	26	Scheduled Date/Time
37	Number of Sample Containers	4	Number of Sample Containers
38	Transport Logistics of Collected Sample	60	Transport Logistics of Collected Sample
39	Collector's Comment	200	Result Status
40	Transport Arrangement Responsibility	60	Parent Result
41	Transport Arranged	30	Quantity/Timing
42	Escort Required	1	Result Copies To
43	Planned Patient Transport Comment	200	Parent
44	Ordering Facility Name	60	Transportation Mode
45	Ordering Facility Address	106	Reason for Study
46	Ordering Facility Phone Number	48	Principal Result Interpreter
47	Ordering Provider Address	106	Assistant Result Interpreter

Note: this message is only used to ORU ^ R01 message. Fields 1, 11, and 37 use integer data types; All other fields use string data types. 12, 13, 14, 15, 17, 18, 19, 20 field according to the number of quality control liquid the format are V1 ^ ^ V2...Vi. It should be noted that for daytime quality control, each quality control test constitutes a message sending; The real-time and intraday quality control is a message sent by several quality control tests of each project together, but different batches of quality control tests are sent separately.

2.3.5 OBX – Observation

OBX is mainly used to deliver observed information in report messages. If the patient's sample test information is transmitted (when msh-16 is 0) - a patient may have multiple OBX, this interface does not provide repeated detection of the test data, and the data is processed by the client system itself.

YHLO HL7 interface OBX use below fields:

No.	Field	Length	Description
1	Set ID – OBX	10	Set ID – OBX
2	Value Type	3	Value Type NM (numeric) ST (string) BOTH: There are both quantitative and qualitative results
3#	Observation Identifier	590	Check the project channel number. In the operating software, set the Lis channel number of the project in the interface for the project parameters, which can be Numbers or letters The test results sent by LIS receiving instrument must be matched by this field
4	Observation Sub-ID	20	Observation Sub-ID
5	Observation Value	65536	Observation Value
6	Units	90	Units
7	References Range	90	References Range
8	Abnormal Flags	5	Abnormal Flags L-low H-high N-normal
9	Probability	5	Possibility, used as qualitative test result value (negative (-), positive (+),

			weak positive (+-), etc.)
10	Nature of Abnormal Test	2	Abnormal test causes used as a qualitative reference value
11#	Observe Result Status	1	Observe the status of the results and take f-final results
12	Date Last Observe Normal Values	26	Leave empty.Finally, observe the normal date
13	User Defined Access Checks	20	User custom access checks used as raw results
14	Date/Time of the Observation	28	Observe the date/time for checking time
15	Producer's ID	60	Leave empty.Used as inspection department
16	Responsible Observer	80	An observer, used as a medical examiner
17	Observation Method	60	Leave empty.Observation method

Note: this message appear only in the ORU ^ R01 message.Fields 1, 3, 9 use integer data types;Fields 5 and 13 use floating point data types;All other fields use string data types.Need to be aware that as the SI program testing, 5, 13 field format for results 1 ^ 2 ^ 3 results, including 1 results for turbidity (L), the results of 2 as hemolysis (H), the results of 3 for jaundice (I), SI project of the three results respectively.

2.3.6 ORD – Query Definition Segment

YHLO HL7 interface ORD use below fields:

No.	Field	Length	Description
1#	Query Date/Time	26	The query generation time, the system time
2#	Query Format Code	1	Query the format code, and select R (record-oriented format)
3#	Query Priority	1	Query priority, D (deferred)
4#	Query ID	10	Query ID, representing different queries, increases with the number of queries by 1

5	Deferred Response Type	1	Leave empty.Delayed response type
6	Deferred Response Date/Time	26	Leave empty.Delay response date/time
7#	Quantity Limited Request	10	Quantity limit required, take RD (Records)
8#	Who Subject Filter	60	Query the person filter, used as a sample bar code for the patient
9#	What Subject Filter	60	Query content filter, query time set as OTH
10#	What Department Data Code	60	Leave empty.Department data code
11	What Data Code Value Qual.	20	Leave empty.Data code value qualification
12	Query Results Level	1	Leave empty.Query result level, take T (Full results)

Note: this message may appear in QRY ^ Q02 messages and DSR ^ Q03 message.Field 8 is numbered for the sample bar code when it is acquired in real time and empty when it is acquired in batch or by number.Field 4 is an integer data type;All other fields use string data types.

2.3.7 QRF – Query Filter Segment

The QRF segment and QRD are used together to further refine the original query content.The QRF segment of the YHLO HL7 interface USES the following domains:

No.	Field	Length	Description
1#	Where Subject Filter	20	Inquirer location filter, pick device model
2	When Data Start Date/Time	26	Record the start date/time as the beginning of the sample collection time at query time
3	When Data End Date/Time	26	Record the end date/time as the end of the sample collection time at query time
4	What User Qualifier	60	Sample start number, string

5	Other QRY Subject Filter	60	Sample termination number, string.When querying a single sample, the number is the same as the start number
6	Which Date/Time Qualifier	12	RCT (Specimen receipt date/time, receipt of specimen in filling ancillary (Lab))
7	Which Date/Time Status Qualifier	12	COR (Corrected only (no final with corrections))
8	Date/Time Selection Qualifier	12	ALL (All values within the range)
9	When Quantity/Timing Qualifier	60	Leave empty.Interval segment

Note: the message appear in the QRY ^ Q02 and DSR ^ Q03 message.Fields 2 and 3 are the zero of the query date and the time when the query occurs, respectively, and are used as time interval conditions for the query.All fields use string types.Fields 4 and 5 are the sample start number and sample end number of the query respectively, which can satisfy the query by number interval.When field 4 is empty, use time query; when field 4 is not empty, query by sample number.

2.3.8 ERR – Error Segment

The ERR segment is used to add error description in the confirmation message, and the ERR segment of the YHLO HL7 interface USES the following fields:

No.	Field	Length	Description
1#	Error Code and Location	80	Error code and location

Note: this message may be QCK ^ Q02, DSR ^ ^ Q03 or an ACK Q03 message (specific content see message error status code table).This message segment has only one field and USES an integer data type.

2.3.9 QAK – Query Acknowledgment Segment

The QAK segment contains some information following the query reply. The QAK segment of the YHLO HL7 interface USES the following fields:

No.	Field	Length	Description
1	Query Tag	32	Query mark, take SR (represents sample application information)
2	Query Response Status	2	OK: Data found, no errors NF: No data found, no errors AE: Application error AR: Application reject

Note: this message will appear on QCK ^ Q02 and DSR ^ Q03 message. All fields use string data types

2.3.10 DSP – Display Data Segment

The DSP segment is used to display the sample application information and patient information obtained by the query, which can be repeated. The DSP segment of the YHLO HL7 interface USES the following fields:

No.	Field	Length	Description
1	Set ID - DSP	4	Identify different DSP segments
2	Display Level	4	Display Level
3#	Data Line	300	Data Line
4	Logical Break Point	2	Logical Break Point
5	Result ID	20	Result ID

Note: this message appear only in the DSR ^ Q03 message. Field 1 USES an integer data type, while all other fields use a string data type.

The third field, "Data Line", is used to display the sample application information downloaded from the LIS server. The detailed entry and sequence of sample application information are shown in the following table. Where the barcode number and item number are required, the rest of the information can be null, and when the sample contains more than one item, the field will be added from field 29 in the same format.

No.	Field	Length	Description
1	admission number	String	

2	Ned number	String	
3	patient's name	String	Up to 32 characters
4	Date of birth	String	YYYYMMDDHHmmSS Such as: 20160101235959
5	Gender	String	Male = M; Female = F; Other= O
6	Blood type	String	O、A、B、AB
7	Race	String	
8	Address	String	
9	Postcode	String	
10	Home telephone	String	
11	Sample position	String	Sample frame number ^ rack sample a number of such as: 2 ^ 1 (2, 1 cup)
12	Sample collection time	String	YYYYMMDDHHmmSS
13	unused		
14	unused		
15	The patient category	String	outpatient; inpatient; Other
16	Medical insurance account	String	
17	Charge type	String	own; insurance
18	national	String	
19	Native place	String	
20	countries	String	

21#	The sample code	String	
22#	Sample number	int	
23	Inspection time	String	YYYYMMDDHHmmSS
24#	Whether the emergency	String	Y; N; N
25	unused	String	
26#	Sample type	String	Serum; Plasma; Urine; Cerebrospinal fluid; other
27	Doctor	String	
28	inspection department	String	
29#	Test project	string	Project channel number ^ project name ^^ units reference range For example: 100 ALT ^^ 10.1 20.5 g/ml ^ In addition to project channel number and all the other three can be empty, for example 100 ^^ ^

For the same project (such as ALT), if the channel number of the project set on the instrument is 100, LIS must use the same channel number (100) to transmit the sample information. The channel number of the test project of the instrument can be set in [application]->[system]->[LIS setting] for editing.

2.3.11 DSC – Continuation Pointer Segment

DSC - Continuation pointer segment

No.	Field	Length	Description
1	Continuation pointer	180	Continuation pointer

Note: this message is used only for the DSR ^ Q03 message. In DSR ^ Q03 message to reply batch query, except the last one sample application information DSR message in which only the value of the field is empty (as a symbol of data transfer is completed), the rest of the DSR message in this field not only is empty. The message segment has only one field, and its data type USES an integer type.

For example, LIS needs to send 5 sample information in batches:

The first sample DSC, |, 1, |

Second sample DSC, |, 2, |

Third sample DSC, |, 3, |

Sample 4: DSC, |, 4, |

Fifth sample DSC, |, |

2.4 Example

Note: in the following example, the project channel number USES the project English name

2.4.1 Test results upload to LIS server

2.4.1.1 Sample test results

Sample results were sent, including 3 project results: TP: qualitative project, UREA: quantitative and qualitative project, ALT: quantitative project

Instrument transmission: ORU^R01

```
<SB>MSH|^~\&|YHLO|iFlash3000|||20160122105550||ORU^R01|9|P|2.3.1|||0||ASCII|||<CR>  
>
```

```
PID|1|HosNo1|MedicalRecordSN1|BedNo1|name1|Area1|19800101000000|F|B||Address1|ZipCode1|TelNo1|||PatientType1|SecuritySN1|FeeType1|AnimalType1|Nation1|Native1||Remark1|Country1|||<CR>
```

```
OBR|1|BarCode1|1|YHLO^iFlash3000|N|20160122094834|20160122094834||1.000000|1^1||Diagnosis1|20160122094834|SampleType1|RequestDoctor1|RequestDept1|sSampleAttr1|BloodBagNo1|TreatDoctor1|TreatDept1|||<CR>
```

```
OBX|1|ST|TP|TP||ResultUnit1|||+|RefValue1|F||2.470000|20160122094834|TestDept1|TestDoctor1|<CR>
```

```
OBX|2|BOTH|UREA|UREA|3.230000|ResultUnit2|6.560000|L|+|-|RefValue2|F||20160122094834|TestDept2|TestDoctor2||<CR>
```

```
OBX|3|NM|ALT|ALT|4.230000|ResultUnit3|7.560000-10.890000|H||F||4.470000|20160122094834|TestDept3|TestDoctor3||<CR>
```

```
<EB><CR>
```

■ **LIS reply:** ACK^R01

```
<SB>MSH|^~\&||YHLO|iFlash3000|20160122105550||ACK^R01|9|P|2.3.1|||0||ASCII|||<CR>
```

```
MSA|AA|9|Message accepted|||0|<CR>
```

```
<EB><CR>
```

2.4.1.2 Quality control test results

Send quality control results of three projects, namely ALT, TP and UREA, each project contains multiple quality control products and multiple test results of multiple batches

■ instrument sent: ORU ^ R01, the first project

```
<SB>MSH|^~\&|YHLO|iFlash3000|||20160122110035||ORU^R01|10|P|2.3.1|||2||ASCII|||<CR>
OBR|9|ALT|ALT|YHLO|iFlash3000|||20160122094834||0|2|3|QCNo^QCNo^QCNo|QCName^QC
Name^QCName|LotNo^LotNo^LotNo|20160221094834^20160221094834^20160221094834||L^
M^H|1.230000^2.230000^3.230000|4.560000^5.560000^6.560000|7.890000^8.890000^9.890000|
|||||||||||||||||<CR>
<EB><CR>
```

■ LIS reply: ACK^R01

```
<SB>MSH|^~\&|YHLO|iFlash3000|20160122110035||ACK^R01|10|P|2.3.1|||2||ASCII|||<CR>
MSA|AA|10|Message accepted|||0|<CR>
<EB><CR>
```

■ instrument sent: ORU^R01 , the second project

```
<SB>MSH|^~\&|YHLO|iFlash3000|||20160122110036||ORU^R01|11|P|2.3.1|||2||ASCII|||<CR>
OBR|10|TP|TP|YHLO|iFlash3000|||20160122094834||0|2|3|QCNo^QCNo^QCNo|QCName^QCN
ame^QCName|LotNo^LotNo^LotNo|20160221094834^20160221094834^20160221094834||L^M
^H|1.230000^2.230000^3.230000|4.560000^5.560000^6.560000|7.890000^8.890000^9.890000|||
|||||||||||||||||<CR>
<EB><CR>
```

■ LIS reply: ACK^R01

```
<SB>MSH|^~\&|YHLO|iFlash3000|20160122110036||ACK^R01|11|P|2.3.1|||2||ASCII|||<CR>
MSA|AA|11|Message accepted|||0|<CR>
<EB><CR>
```

■ instrument sent: ORU^R01 , the third project

```
<SB>MSH|^~\&|YHLO|iFlash3000|||20160122110037||ORU^R01|12|P|2.3.1|||2||ASCII|||<CR>
OBR|11|UREA|UREA|YHLO|iFlash3000|||20160122094834||0|2|3|QCNo^QCNo^QCNo|QCName^QCName^QCName|LotNo^LotNo^LotNo|20160221094834^20160221094834^20160221094834||L^M^H|1.230000^2.230000^3.230000|4.560000^5.560000^6.560000|7.890000^8.890000^9.890000|||
|||||||||||||||||<CR>
<EB><CR>
```

■ LIS reply: ACK^R01

```
<SB>MSH|^~\&|||YHLO|iFlash3000|20160122110037||ACK^R01|12|P|2.3.1|||2||ASCII|||<CR>  
MSA|AA|12|Message accepted|||0|<CR>  
<EB><CR>
```

2.4.2 Obtain sample information from LIS server

2.4.2.1 Search by barcode

The query bar code is a sample of BarCode

■ instrument sent: QRY^Q02

```
<SB>MSH|^~\&|||YHLO|iFlash3000|||20160122110540||QRY^Q02|14|P|2.3.1|||ASCII|||<CR>  
QRD|20160122110540|R|D|1||RD|BarCode1|OTH||T|<CR>  
QRF|iFlash3000|||RCT|COR|ALL||<CR>  
<EB><CR>
```

■ LIS reply: QCK^Q02

```
<SB>MSH|^~\&|||YHLO|iFlash3000|20160122110540||QCK^Q02|14|P|2.3.1|||ASCII|||<CR>  
MSA|AA|14|Message accepted|||0|<CR>  
ERR|0|<CR>  
QAK|SR|OK|<CR>  
<EB><CR>
```

■ LIS reply: DSR^Q03, the first project

```
<SB>MSH|^~\&|||20160122110540||DSR^Q03|14|P|2.3.1|||ASCII|||<CR>  
MSA|AA|14|Message accepted|||0|<CR>  
ERR|0|<CR>  
QAK|SR|OK|<CR>  
QRD|20160122110540|R|D|1||RD||OTH||T|<CR>  
QRF|iFlash3000|||RCT|COR|ALL||<CR>  
DSP|1||AD000|||<CR>  
DSP|2||bed NO.000|||<CR>  
DSP|3||name000|||<CR>  
DSP|4||20000101000000|||<CR>  
DSP|5||M|||<CR>
```

DSP|6||blood type000|||<CR>
 DSP|7|||||<CR>
 DSP|8||address000|||<CR>
 DSP|9||postcode0|||<CR>
 DSP|10||phone0|||<CR>
 DSP|11||1^1|||<CR>
 DSP|12|||||<CR>
 DSP|13|||||<CR>
 DSP|14|||||<CR>
 DSP|15|| The patient category 000|||<CR>
 DSP|16|| Medical insurance account 000|||<CR>
 DSP|17|| Charge type 000|||<CR>
 DSP|18|| national 000|||<CR>
 DSP|19|native 000|||<CR>
 DSP|20|| countries 000|||<CR>
 DSP|21||BarCode1|||<CR>
 DSP|22||1|||<CR>
 DSP|23|||||<CR>
 DSP|24||N|||<CR>
 DSP|25|||||<CR>
 DSP|26|| Sample type 0|||<CR>
 DSP|27|Doctor0|||<CR>
 DSP|28|| The inspection department 0|||<CR>
 DSP|29||ALT^ALT^result unit0^|||<CR>
 DSP|30||TP^TP^ result unit 1^|||<CR>
 DSP|31||UREA^UREA^ result unit 2^|||<CR>
 DSC|1|<CR>
 <EB><CR>

- **Instrument sent:** ACK^Q03, Receive confirmation of the first sample information

■ <SB>MSH|^~\&|YHLO|iFlash3000|||20160122110541||ACK^Q03|14|P|2.3.1|||||ASCII|||<CR>
 >
 MSA|AA|14|Message accepted|||0|<CR>
 ERR|0|<CR>
 <EB><CR>

- **LIS reply:** DSR^Q03, The second sample information (note: it is assumed that there are multiple samples that meet the query

conditions)

```
<SB>MSH|^~\&||||20160122110540||DSR^Q03|14|P|2.3.1|||||ASCII|||<CR>
MSA|AA|14|Message accepted|||0|<CR>
ERR|0|<CR>
QAK|SR|OK|<CR>
QRD|20160122110540|R|D|1||RD||OTH||T|<CR>
QRF|iFlash3000||||RCT|COR|ALL||<CR>
DSP|1||HosNo1|||<CR>
DSP|2||BedNo1|||<CR>
DSP|3||name1|||<CR>
DSP|4||19800101000000|||<CR>
DSP|5||F|||<CR>
DSP|6||B|||<CR>
DSP|7|||||<CR>
DSP|8||Address1|||<CR>
DSP|9||ZipCode1|||<CR>
DSP|10||TelNo1|||<CR>
DSP|11||1^1|||<CR>
DSP|12|||||<CR>
DSP|13|||||<CR>
DSP|14|||||<CR>
DSP|15||PatientType1|||<CR>
DSP|16||SecuritySN1|||<CR>
DSP|17||FeeType1|||<CR>
DSP|18||Nation1|||<CR>
DSP|19||Native1|||<CR>
DSP|20||Country1|||<CR>
DSP|21||BarCode1|||<CR>
DSP|22||1|||<CR>
DSP|23|||||<CR>
DSP|24||N|||<CR>
DSP|25|||||<CR>
DSP|26||SampleType1|||<CR>
DSP|27||RequestDoctor1|||<CR>
DSP|28||RequestDept1|||<CR>
DSP|29||TP^TP^ResultUnit1^|||<CR>
DSP|30||UREA^UREA^ResultUnit2^|||<CR>
DSP|31||ALT^ALT^ResultUnit3^|||<CR>
DSC||<CR>
<EB><CR>
```

■ **Instrument sent:** ACK^Q03, Receive confirmation of the second

sample information

```
<SB>MSH|^~\&|YHLO|iFlash3000|||20160122110542||ACK^Q03|14|P|2.3.1|||||ASCII|||<CR>  
MSA|AA|14|Message accepted|||0|<CR>  
ERR|0|<CR>  
<EB><CR>
```

2.4.2.2 Query by sample number

Query the samples for sample Numbers 1 through 9

■ Instrument sent: QRY^Q02

```
<SB>MSH|^~\&|YHLO|iFlash3000|||20160122112214||QRY^Q02|15|P|2.3.1|||||ASCII|||<CR>  
QRD|20160122112214|R|D|1||RD||OTH||T|<CR>  
QRF|iFlash3000||19|RCT|COR|ALL||<CR>  
<EB><CR>
```

The rest of the procedure with bar code mode query.

2.4.2.3 Query by time

Query the samples of 20160122 from 8:00 to 12:00

■ Instrument sent: QRY^Q02

```
<SB>MSH|^~\&|YHLO|iFlash3000|||20160122112815||QRY^Q02|16|P|2.3.1|||||ASCII|||<CR>  
QRD|20160122112815|R|D|1||RD||OTH||T|<CR>  
QRF|iFlash3000|20160122080000|20160122120000||RCT|COR|ALL||<CR>  
<EB><CR>
```

The rest of the procedure with bar code mode query.

2.4.3 LIS server sends sample information actively

The LIS server actively sends a sample message

■ LIS sent: DSR^Q03

```
<SB>MSH|^~\&|||||20160122113108||DSR^Q03|4|P|2.3.1||P|||ASCII|||<CR>  
MSA|AA|4|Message accepted|||0|<CR>
```

ERR|0|<CR>
 QAK|SR|OK|<CR>
 QRD|20160122113108|R|D|4||RD||OTH|||T|<CR>
 QRF|iFlash3000||||RCT|COR|ALL||<CR>
 DSP|1||AD000|||<CR>
 DSP|2||bed NO.000|||<CR>
 DSP|3||name000|||<CR>
 DSP|4||20000101000000|||<CR>
 DSP|5||M|||<CR>
 DSP|6||bllod type000|||<CR>
 DSP|7|||||<CR>
 DSP|8||address000|||<CR>
 DSP|9||postcard0|||<CR>
 DSP|10||phone0|||<CR>
 DSP|11||1^1|||<CR>
 DSP|12|||||<CR>
 DSP|13|||||<CR>
 DSP|14|||||<CR>
 DSP|15|| patient category 000|||<CR>
 DSP|16|| Medical insurance account 000|||<CR>
 DSP|17|| Charge type 000|||<CR>
 DSP|18|| national 000|||<CR>
 DSP|19|| Native place 000|||<CR>
 DSP|20|| countries 000|||<CR>
 DSP|21||BarCode1|||<CR>
 DSP|22||1|||<CR>
 DSP|23|||||<CR>
 DSP|24||N|||<CR>
 DSP|25|||||<CR>
 DSP|26|| Sample type 0|||<CR>
 DSP|27||Doctor0|||<CR>
 DSP|28|| The inspection department 0|||<CR>
 DSP|29||ALT^^^|||<CR>
 DSP|30||TP^^^|||<CR>
 DSP|31||UREA^^^|||<CR>
 DSC||<CR>
 <EB><CR>

■ Instrument reply: ACK^Q03

<SB>MSH|^~\&|YHLO|iFlash3000||20160122113108||ACK^Q03|4|P|2.3.1|||||ASCII|||<CR>
 MSA|AA|4|Message accepted|||0|<CR>
 ERR|0|<CR>
 <EB><CR>

3.ASTM Protocol

3.1. Overview

3.1.1. Protocol Version

This interface is based on the version of ASTM e1394-97. Since this agreement only USES part of ASTM protocol, only the part used is described later.

3.1.2. Underlying protocol

The ASTM protocol link layer works in simplex mode. When the instrument sends data frames, the LIS system can only wait for receiving, and when the LIS system sends data frames, the instrument can only wait for receiving. Therefore, the ASTM underlying protocol states that before any party sends data, it is necessary to obtain the send authority.

3.1.2.1. Get the data frame transmission right of link

ASTM protocol stipulates that when the sender needs to send a data frame, it should first send the <ENQ> control command character to the receiver. If the receiver is ready to receive, the reply <ACK> to the sender; otherwise, the reply <NAK>.

When the sender receives the reply <ACK> of <ENQ>, the sender obtains the data frame transmission right of the link and can send the data frame. When the data frame transmission is completed, the control command character of <EOT> is sent to the receiver and the sending process ends.

When the sender receives the response <NAK> of <ENQ>, it means that the receiver is not ready to receive, at this time, the sender should wait for 10s and send <ENQ> again.

3.1.2.2. Conflict handling of request sending rights

If both parties want to send data at the same time, there will be a conflict, for which the ASTM agreement provides:

(1) after sending <ENQ> control command character, receive <ENQ> control command character. LIS system must stop any attempt of data transmission and be ready to receive data. When the next

<ENQ> control command character is received, reply <ACK> or <NAK> to confirm the receiving readiness.

(2) after sending <ENQ> control command character, receiving <ENQ> control command character, the instrument will send <NAK> and wait for ACK of LIS system. If not, it will continue to send another <ENQ> control command character.

3.1.2.3. Data frame transmission rights and timeout processing of the release link

The party that has received the sending authority sends the <EOT> control command character to inform the receiver that the sending is complete and to release the sending authority of the link after the data frame is sent.

In order to prevent the sender from being released after gaining control, the agreement states that when the sender gains control, no data frames are sent within 15s, and the receiver acquires the right of sending by default.

During data transmission, if the sender sends more than 15s between two bytes, the receiver considers the sender to have timed out, and the receiver will exit the receiving state.

After sending the last byte of the data frame, the sender does not receive the <ACK> of the receiver within 15s, so it is considered that the receiver has timed out, and the sender exits the sending state.

3.1.3. Get the data frame transmission right of link

In addition to sending <ENQ>, <NAK>, <ACK>, and <EOT> when used for the link layer transmit and receive control, the data of the application layer should be sent in the following format:

Intermediate frame format: <STX> FN < DATA >< ETB> <CS> <CR><LF>

End frame format: <STX> FN < DATA >< ETX><CS> <CR><LF>

Code	Description
<STX>	Start frame header (0x02)
FN	Frame number, ASCII '0' to '7'
< DATA >	The application layer needs to send the data content
<ETB>	Text block end character (0x17)
<ETX>	End of frame (0x03)
<CS>	Check sum, 2 characters

<CR>	Carriage return (0x 0D)
<LF>	Line breaks (0x 0A).

When the application layer sends more data content at one time, it can be divided into multiple data frames for transmission. It is not necessary to divide frames, only end frames are used regardless of the frame.

Considering the sending of serial port, the message quantity of a message cannot be too large. This agreement provides for unpacking and sending of the sent data, and each record is sent as a single frame.

3.1.4. Checksum calculation

Add up each byte of FN to <ETB> or <ETX>, where each byte is an unsigned positive integer (range: 0x00-0xff), and the resulting complement is 0x100(aka 256 in decimal), resulting in a hexadecimal number between 0x00-0xff. Two ASCII symbols are used for transport, such as the character "4F" for accumulation and 0x4F for hexadecimal.

Example: calculate <STX>123abc<ETX> verification

Code	Hexadecimal	Plus
<STX>	0x02	0x00(<STX> does not participate in the calculation)
'1'	0x31	0x31
'2'	0x32	0x31+0x32 = 0x63
'3'	0x33	0x63+0x33 = 0x96
'a'	0x61	0x96+0x61 = 0xF7
'b'	0x62	0xF7+0x62 = 0x159
'c'	0x63	0x159+0x63 = 0x1BC
<ETX>	0x03	0x1BC+0x03 = 0x1BF
	Complementary to 0x100(that is, 256 decimal Numbers)	0x1BF mod 0x100 = 0xBF

Then send <STX>123abc<ETX>BF (note that the hexadecimal number symbol ABCDEF is all in upper case).

3.1.5. Control Code

Obtain the link layer send weight, and the control characters required for data frames are standard ASCII coded controls, including:

Code	Hexadecimal value
<ENQ>	0x05
<ACK>	0x06
<EOT>	0x04
<NAK>	0x15
<STX>	0x02
<LF>	0x0A
<CR>	0x0D
<ETB>	0x17
<ETX>	0x03

3.1.6. The application layer

3.1.6.1. Supported message logging

- 1) Message Header Record
- 2) Patient Information Record
- 3) Test Order Record
- 4) record of test results (Result Record)
- 5) Comment Record
- 6) Request Information Record
- 7) Message Terminator Record

The sent data will contain some or all of the seven message records above.

Code	Description
H	Header Record
P	Patient Record
O	Order Record
R	Result Record
C	Comment Record
Q	Request Information Record (Host query)
L	Terminator Record

3.1.6.2. Message structure

A completed message has multiple message records, and the specific composition of each message is as follows:

The instrument sends the sample results to the Lis server

Message Header Record	(H)
Patient Information Record	(P)
Test Order Record	(O)
{	
Result Record	(R)
Results Comment Record	(C)
}(n)	A test order record can have one or more result
Message Terminator Record	records end of message logging (L)

The instrument sends quality control results to Lis server

Message Header Record	(H)
Patient Information Record	(P)
Test Order Record	(O)
Message Terminator Record	(L)

The instrument enquiries with Lis server for sample information

Message Header Record	(H)
Request Record	(Q)
Message Terminator Record	(L)

Lis server sends sample information to the instrument

The following structure is used for both sending and responding to queries

Message Header Record	(H)
{	
Patient Information Record	(P)
{	
Test Order Record	(O)
}(n)	A test order record can have one or more result
}(n)	records end of message logging (L)
Message Terminator Record	

3.1.6.3. Message syntax

Each ASTM message consists of a number of message records that end with <CR>.

Each record consists of a 1-character record name and a number of fields, which are made up of components, and H records in each message define the delimiters of the constituent units.

Such as:

H | \ ^ & || iFlash3000 ^ 01.00.00.03 ^ 123456 || || || || PR 1394-97 | 20180101235959 <CR>

Where, the four characters after H define the delimiter used to distinguish between fields, repetitions, and components. The YHLO ASTM interface USES the following table characters:

Code	Description
	Field delimiter
\	Repetition separator
^	Component separator
&	Transfer character

When the domain content contains the above table characters, the escape character needs to be used.

3.1.6.4. Code transfer

In string type field data, such as remarks, diagnostic information, user-defined gender and other string data, escape delimiters may appear. The delimiters in the original string should be escaped into an escape character sequence when encoding, and then restored when decoding. The escape rule is as follows:

Code	The original character
&F&	Field separator ' '
&R&	Repetition separator '\'
&S&	The component separator '^'
&E&	Escape character '&'

3.2. Message record definition

3.2.1. HeaderRecord- “H”

Any message is sent as a message header with an "H" message record. In the second domain of an H message, the delimiter is required in the message, so each message must contain an H message header

No.	The field name	content	Lenth	Description
1#	Record Type ID	H	1	Record type ID
2#	Delimiter Definition	 \ ^ &	4	Field separator
3	Message Control ID		-	Repetition separator
4	Password		-	Component separator
5	Sender Name or ID	Character set coding	40	Escape character
	Sowftware Version	PR (patient test results)	20	Leave empty.
	Serial Number	QR (quality control test results)	20	Leave empty.
6	Sender Street Address	CR (calibration results)	-	The name of the instrument
7	Reserved field	RQ (sample request query)	-	Software version
8	Sender Telephone Number	QA (sample query response)	-	Instrument serial number
9	Characteristics of sender	SA (sample application information)	-	Leave empty.
10	Receiver ID	Supports strings and Numbers	-	Leave empty.
11	Comment or Special Instructions	Character set coding	-	Leave empty.
12#	Processing ID	PR (patient test results)	2	Leave empty.
13#	Version Number	QR (quality control test results)	12	Leave empty.
14	Date and Time	YYYYMMDDHHMMSS	14	Character set: ASCII represents the default character encoding of the operating system (generally GBK for WIN7 Chinese operating system), and UTF8 represents UTF8 encoding

Example:

H|\^&|||iFlash3000^01.00.00.00^123456|||||PR|1394-97|20171017191404<CR>

3.2.2. Patient Record – “P”

In obtaining or downloading sample information and sending sample test results information, patient information is required, and it is part of the sample information.

No.	The field name	content	Lenth	Description
1#	Record Type ID	P	1	Record type ID
2	Sequence Number		3	The serial number The sequence number indicating patient information is recorded at the current level, the first patient information record is set to 1 at the same message header, and the second is set to 2, and so on.
3	Practice Assigned Patient ID		–	reserved
4	Patient ID	string	25	Medical record no.
5	Patient ID #3	string	25	In the hospital,
6#	Patient Name			The patient's name
7	Reserved field		–	Leave empty.
8	Birth date	string	8	Date of birth: 20180101
	Age	digital	6	age
	Age Unit	Y = year M = Month W = week D = Days H = hours If empty =Year	1	Age unit
9	Patient Sex	M(female) F(male) U(unknown)	1	Patient sex
10	Patient Race	string	20	national
11	Patient Address	string	50	Patients address
12	Reserved Field	A B O AB Other	20	Used as a blood type
13	Patient Telephone	string	13	contact
14	Attending Physician Name			reserved
15	Special field 1			reserved
16	Body Surface Area (M2)	string	20	AD account
17	Patient Height		–	reserved

No.	The field name	content	Lenth	Description
	Patient Height Unit		–	reserved
18	Patient Weight		–	reserved
	Patient Weight Unit		8	reserved
19	Patient Diagnosis			reserved
20	Patient Medications			reserved
21	Patient Diet	string	20	Blood bag Number
22	Field #1		–	reserved
23	Practice Field #2		–	reserved
24	Admission and dischargedates		–	reserved
25	Admission Status		–	reserved
26	Location	string	10	location
27	Nature of alt. Diag. Code and Class.	string	10	Nature of alt. Diag
28	Alt. Diag. Code and Class		–	reserved
29	Patient Religion		–	reserved
30	Marital Status		–	reserved
31	Isolation Status		–	reserved
32	Language		–	reserved
33	Hospital Service		–	reserved
34	Hospital Institution		–	reserved
35	Dosage Category		–	reserved

for the p-6 patient Name, the user can directly fill in the Last Name.

For example, Lis sends patient information:

```
H|\^&||iFlash3000^01.00.00.00^123456|||||SA|1394-97|20180101235959<CR>
P|1||PATIENT111||Smith^Tom^J||19600315|M|||A||Dr. Bean|icteru|100012546|||Diagnosis
information||0001||||A1|002|||||||<CR>
L|1|N<CR>
```

3.2.3. Test Order Record – “O”

The test sheet records the information related to the test report, including regular sample and quality control sample, which are described below.

3.2.3.1. Sample results

The following domains were used in the YHLO ASTM interface when the patient sample test results were transmitted (PR/SA/QA was taken from h-12)

No.	The field name	content	Lenth	Description
1#	Record Type ID	0	1	Record Type ID
2	Sequence Number	Number	3	Sequence Number indicates the Sequence Number of the test order recorded in the current layer, the first test order record is set to 1 and the second is set to 2 in the same patient information, and so on.
3	Sample ID	stringNumber	20	Sample number
	Sample Tray No.	Number	2	Sample frame number/sample plate number (this field will be ignored when LIS is issued)
	Sample Pos	Number	2	The location of the sample on the rack/disk (when LIS is issued, the field will be ignored by the instrument)
4	Instrument SpecimenID	string	29	The sample code
5	Assay No.	string	12	Item channel number
	Assay Name	string	20	The project name
	Dilution Rate	Number	4	Dilution times, 10 means the dilution rate is 1:10
	Repeat Num	Number	2	Leave empty
	Rerun flag	Number	1	Retest mark: 0 for non-retest 1 for retest
6	Priority	R(Normal) S(STAT)	1	LIS server is sent to the instrument, the symbol for 1 project need to remeasure, flag is 0 means projects do not need to retest, the flag is 1 of the project, the instrument side no matter how many times the project has completed the test, all will be carried out to test, for signs of 0 project, if the project has been completed before testing, will not test again. Through this symbol, the retest function can be realized.
7	Requested Date and Time (Sample program creation date)	string	14	Whether the emergency
8	Specimen Collection Date and Time	string	14	Sample application time
9	Collection end Time			20180101235959

No.	The field name	content	Lenth	Description
10	Collection Volume	0-99999.0 Max = 99999.0	7	Sample collection time
11	Collected by (ID or Name)	string	18	20180101235959
12	Action Code	string	-	reserved
13	Danger Code	string	-	Collection capacity
14	Relevant Clinical Information	string	-	gatherers
15	Date/Time Specimen Received in the Lab.	string	14	reserved
16	Specimen type (This field is not translated to local languages)	serum urine CSF plasma timed other blood amniotic urethral saliva cervical synovial	10	reserved
17	Ordering Physician			Send the doctor
18	Physician Phone number	string	30	The inspection department
19	Offline Dilution factor	Number	4	Manual dilution factor
20	User Field #2			Check the doctor
21	Laboratory field #1		-	reserved
22	Laboratory field #2		-	reserved
23	Date/Time Results Reported/Mod		-	reserved
24	Instrument Charge to computer system		-	reserved
25	Instrument Section ID		-	reserved
26#	Report Type	0 (request from LIS) Q (query response) F (final result)	1	The report type
27	Reserved Field		-	reserved
28	Location Specimen Collected		-	reserved
29	Nosocomial Infection		-	reserved

No.	The field name	content	Lenth	Description
	flag			
30	Specimen Service		-	reserved
31	Specimen Institution		-	reserved

The project code in field 5 is the project code that can be recognized by both parties. Use the repetition separator "\" between multiple items, up to 160 items, over 160 items, need to be transmitted separately;

When the instrument sends the sample test results to the LIS system, the ReportType value of the 26th field is F. When the LIS system sends sample information responding to the instrument query request, the ReportType value of the 26th field is Q. When LIS system actively sends sample information to the instrument, the ReportType value of field 26 is O.

For example, a sample application issued by Lis:

H | \ ^ & | | | iFlash3000 ^ 01.00.00.00 ^ 123456 | | | | | SA | | < CR > 20180101235959,
1394-97

P | 1 | | PATIENT111 | | ^ ^ Tom Smith J | | 19600315 | M | | A | | icteru | | | | | s | | | | | A1 |
002 | | | | | | | < CR >

O | | 1 ^ ^ 1 | SAMPLE123 | 1 ^ Test1 ^ 2 ^ 2 ^ 1 \ Test2 ^ 2 1 \ ^ 3 ^ Test3 ^ 2 ^ 1 \ ^ 4 Test4
^ 2 ^ 1 | R | 20180101080000 | 20160103080000 | | John | | | | | Urine | Dr. Who | Department1 | 1 |
Dr. Tom | | | | | O | | | | | < CR >

L | | 1 N < CR >

3.2.3.2. QC results

When transmitting the information of project quality control test results (when h-12 takes QR), the following fields are used in the YHLO ASTM interface

No.	The field name	content	Lenth	Description
1#	Record Type ID	0	1	Record Type ID
2	Sequence Number	Number	3	Sequence Number
3	Sample ID	stringandNumber	20	reserved
	Sample Tray No.	Number	2	reserved
	Sample Pos	Number	2	reserved
4	Instrument SpecimenID	string	29	reserved
5	Assay No.	string	12	Assay No.
	Assay Name	string	20	Assay Name
	Dilution Rate	Number	4	reserved
	Repeat Num	Number	2	reserved
6	Priority	R(normal) S(STAT)	1	reserved
7	Requested Date and	string	14	Requested Date and

No.	The field name	content	Lenth	Description
	Time (Sample program creation date)			20160910121532
8	Specimen Collection Date and Time	string	14	reserved
9	Collection end Time			reserved
10	Collection Volume	0-99999.0 Max = 99999.0	7	reserved
11	Collected by (ID or Name)	string	18	reserved
12	No.	string	10	Quality control liquid number
	Name	string	10	Name of quality control liquid
	Batch No.	string	20	Batch number of quality control liquid
	period of validity	string	8	The quality control liquid is valid for 20090910
	Average Concentration	Number	9	Average of quality control liquid
	Level	H; M; L	1	The concentration level of the quality control solution.
	Standard Diff.	Number	9	Standard deviation of quality control fluid
	Concentration	Number	9	Test result value (concentration)
13	Danger Code	string	-	reserved
14	Relevant Clinical Information	string	-	reserved
15	Date/Time Specimen Received in the Lab.	string	14	reserved
16	Specimen type (This field is not translated to local languages)	serum urine CSF plasma timed other blood amniotic urethral saliva cervical synovial	10	reserved
17	Ordering Physician			reserved
18	Physician Phone number	string	30	reserved
19	Offline Dilution factor	Number	4	reserved
20	User Field #2			reserved

No.	The field name	content	Lenth	Description
	Last Name	string	20	reserved
	First Name	string	20	reserved
	Middle Initial	string	1	reserved
21	Laboratory field #1		–	reserved
22	Laboratory field #2		–	reserved
23	Date/Time Results Reported/Mod		–	reserved
24	Instrument Charge to computer system		–	reserved
25	Instrument Section ID		–	reserved
26	Report Type	0 (request from LIS) Q (query response) F (final result)	1	Report Type
27	Reserved Field		–	reserved
28	Location Specimen Collected		–	reserved
29	Nosocomial Infection flag		–	reserved
30	Specimen Service		–	reserved
31	Specimen Institution		–	reserved

For example, send quality control results to Lis:

```
H|\^&||iFlash3000^01.00.00.00^123456|||||QR|1394-97|20180101235959<CR>
P|1|||||||||||||||||||||||||||<CR>
0|1||||20180101235959||||1^QC1^1111^20100910^10^L^5^10.28\2^QC2^2222^20100910^20^M^10^20.4
8\3^QC3^3333^20100910^30^H^15^30.25|||||||||||F|||||<CR>
L|1|N<CR>
```

3.2.4. Result Record- “R”

The Result Record contains a test Result of a test item, that is to say, a test Result will have a Result Record, which will be included only when a message is sent from the instrument to the LIS service end

No.	The field name	content	Lenth	Description
1#	Record Identifier	R	1	Record Type ID
2	Sequence Number	1-n	3	Sequence Number
3#	Assay No. #	string	12	Assay No. #

No.	The field name	content	Lenth	Description
	Assay Name	string	20	Assay Name
	Replicate number	Number	2	Replicate number
	Result Type	I: qualitative results F: quantitative results B: quantitative results + qualitative description	1	Result Type
4	Measurement Value	string (quantitative)	14	When the type of result is F or B, the concentration value is transmitted. If the result exceeds the upper limit of the instrument's measuring range, the ">" symbol appears before the value. If the result is lower than the lower limit of the instrument's measuring range, the "<" sign appears before the value.
	Interpretation	String (negative (-), positive (+), weak positive (+-), etc.)	15	Qualitative final test results are valid when the type of result is I or B
5	Units	The string	12	Results the unit
6	Measurement Range Upper Limit	Number	12	Upper limit of reference range
	Measurement Range Lower Limit	Number	12	Lower limit of reference range
7	Result Abnormal flag	L(result < reference range)	1	Abnormal node check mark
8	Nature of Abnormality Testing	H(result > reference range)	15	Qualitative reference value, valid when the result type is I or B
9#	Result Status	N (normal)	1	Results:
10				reserved
11	Operator Identification		-	reserved
12	Date/time test started	string	14	reserved
13	Date/Time test Completed	string	14	Test completion time

No.	The field name	content	Lenth	Description
14	Instrument Identification (Sender Name)	string	-	Which module is tested in the cascade product, such as M1.

For example, send sample test results to Lis:

```
H|\^&||iFlash3000^01.00.00.00^123456|||||PR|1394-97|20180101235959<CR>
P|1|PATIENT111|Smith^Tom^J|19600315|M||A||icteru|||||01|||||A1|002||||||<CR>
O|1|1^1^1|SAMPLE123|1^Test1^2^1|2^Test2^2^1|3^Test3^2^1|4^Test4^2^1|R|20090910135300|2009091
0125300|||John|||Urine|Dr. Who|Department1|1|Dr. Tom|||||F||||<CR>
R|1|1^Test1^1^F|14.5^Mg/ml||5.6^99.9|N|F|||20180101235959|M1<CR>
R|2|2^Test2^1^F|3.5^Mg/ml||5.6^50.9|L|F|||20180101235959|M1<CR>
R|3|3^Test3^1^F|24.5^Mg/ml||1.1^20.9|H|F|||20180101235959|M1<CR>
R|4|4^Test4^1^I|^Negative|Mg/ml|||Positive|F|||20180101235959|M1<CR>
L|1|N<CR>
```

3.2.5. Comment Record – “C”

Currently, only the Result Comment is applied, and the description used to represent the test Result in the Result Record is used

No.	The field name	content	Lenth	Description
1#	Record Identifier	C	1	Record Type ID
2	Sequence Number	1-n	3	Sequence Number
3	Comment Source	I: indicates that the analyzer is sent	1	Comment source, fixed value
4	Comment Text	L: stands for LIS host sending	90	Annotation text
5	Comment Type	The string	1	The annotation type

For example, send notes to Lis:

```
H|\^&||iFlash3000^01.00.00.00^123456|||||PR|1394-97|20180101235959<CR>
P|1|PATIENT111|Smith^Tom^J|19600315|M||A||icteru|||||01|||||A1|002||||||<CR>
O|1|1^1^1|SAMPLE123|1^Test1^2^1|R|20180101235959|20180101235959|||John|||Urine|Dr. Who|Depa
rtment1|1|Dr. Tom|||||F||||<CR>
R|1|1^Test1^1^F|14.5^Mg/ml||5.6^99.9|N|F|||20180101235959|<CR>
C|1|I|Result Description|I<CR>
L|1|N<CR>
```

3.2.6. Request Record- “Q”

The Request Record Record is used to query the sample application information to the LIS

server. It is a message with specified patient ID, sample ID, time range and other query conditions sent by the instrument end to the LIS service end. If the LIS service does not return the information of the query within the specified time, the instrument end can send the message of canceling the query to cancel the last query operation.

No.	The field name	content	Lenth	Description
1#	Record Identifier	Q	1	Record Type ID
2	Sequence Number	1-n	3	Sequence Number
3	Patient ID	string	20	Sample start number
	Specimen ID	string	29	The sample code
4	Ending Range ID	string	20	The end of the sample number, when a single sample number is queried, is the same as the start of the sample number
5	Universal Test Id	ALL	-	The system always requests all pending orders.
6	Nature of Request Time Limits		-	reserved
7	Beginning Request results date/time.	string	14	Query start time
8	Ending Request results date/time	string	14	Query cut-off time
9	Requesting Physician name		-	reserved
10	Requesting Physician Telephone		-	reserved
11	User field # 1		-	reserved
12	User field # 2		-	reserved
13#	Request Information status Codes	0: request a sample query A: cancel the current query request	1	Query command code

Example

Contact the Lis:

```
H|\^&||iFlash3000^01.00.00.00^123456|||||RQ|1394-97|20180101235959<CR>
Q|1|^SAMPLE123|||||||0<CR>
L|1|N<CR>
```

Lis returns the record of the sample information queried

```
H|\^&||iFlash3000^01.00.00.00^123456|||||SA|1394-97|20180101235959<CR>
P|1|PATIENT111|Smith^Tom^J|19600315|M||A||icteru|||||01|||||A1|002||||||<CR>
0|1|SAMPLE123^1^1||1^Test1^2^1\2^Test2^2^1\3^Test3^2^1\4^Test4^2^1|R|20180101235959|20180101
235959||||John||||Urine|Dr. Who|Department1|1|Dr. Tom|||||Q|||||<CR>
L|1|N<CR>
```

3.2.7. Terminator Record-“L”

The Terminator Record Record indicates the end of the message at the end of each message.

No.	The field name	content	Lenth	Description
1#	Record Identifier	L	1	Record Type ID
2	Sequence Number	1-n	3	Sequence Number
3#	Terminator code	N=normal termination I=No information available from the last query. Q=Error for in last request for information	1	End code

For example, send end records

```
H|\^&||iFlash3000^01.00.00.00^123456|||||PR|1394-97|20180101235959<CR>
L|1|N<CR>
```

3.3. Communication process and message example

Note: each record in the message is sent separately.

3.3.1. Send sample test results

Send	Content
Instrument	<ENQ>
LIS	<ACK>
Instrument	<STX>1H \^& iFlash3000^01.00.00.00^123456 ASCII PR 1394-97 20180120110408<CR><ETX>A6<CR><LF>

LIS	<ACK>
Instrument	<STX>2P 1 MedicalRecordSN1 HosNo1 zhangsang 19800101^0^Y F Nation1 Address1 B TelNo1 SecuritySN1 PatientType1 Area1 BedNo1 <CR><ETX>CA<CR><LF>
LIS	<ACK>
Instrument	<STX>3O 1 1^1&S&1 BarCode1 TP^TP^1.000000^^0\UREA^UREA^1.000000^^0\ALT^ALT^1.000000^^0 R 20170101000000 20180120105109 1.0 20180120105109 SampleType1 RequestDoctor1 RequestDept1 1.000000 F <CR><ETX>5C<CR><LF>
LIS	<ACK>
Instrument	<STX>4R 1 TP^TP1I ^+ ResultUnit1 + F 20171019114059 <CR><ETX>46<CR><LF>
LIS	<ACK>
Instrument	<STX>5R 2 UREA^UREA1B 3.23^+- ResultUnit2 6.560000^ L +- F 20171019114059 <CR><ETX>AD<CR><LF>
LIS	<ACK>
Instrument	<STX>6R 3 ALT^ALT1F 4.23^ ResultUnit3 7.560000^10.890000 H F 20171019114059 <CR><ETX>2A<CR><LF>
LIS	<ACK>
Instrument	<STX>7L 1 N<CR><ETB>1E<CR><LF>
LIS	<ACK>
Instrument	<EOT>

3.3.2. Send quality control test results

Send	Content
Instrument	<ENQ>
LIS	<ACK>
Instrument	<STX>1H ^& ^^ QR 1394-97 20171019154939<CR><ETX>56<CR><LF>
LIS	<ACK>
Instrument	<STX>2P 1 <CR><ETX>3B<CR><LF>
LIS	<ACK>
Instrument	<STX>3O 1 ALT^ALT 20171019134644 QCNo^QCName^LotNo^20171118^1.230000^L^4.560000^0^7.890000^QCNo^QCName^LotNo^20171118^2.230000^M^5.560000^8.890000^QCNo^QCName^LotNo^20171118^3.230000^H^6.560000^9.890000 F <CR><ETX>7E<CR><LF>
LIS	<ACK>

Instrument	<STX>4L 1 N<CR><ETB>1B<CR><LF>
LIS	<ACK>
Instrument	<EOT>

3.3.3. Send sample query

3.3.3.1. Barcode query

Instrument query

Send	Content
Instrument	<ENQ>
LIS	<ACK>
Instrument	<STX>1H V& ^^ RQ 1394-97 20171019152254<CR><ETX>4A<CR><LF>
LIS	<ACK>
Instrument	<STX>2Q 1 BarCode1 ALL O<CR><ETX>DB<CR><LF>
LIS	<ACK>
Instrument	<STX>3L 1 N<CR><ETB>1A<CR><LF>
LIS	<ACK>
Instrument	<EOT>

LIS system returns the query result

Send	Content
LIS	<ENQ>
Instrument	<ACK>
LIS	<STX>1H V& ^^ QA 1394-97 20171019152254<CR><ETX>39<CR><LF>

Instrument	<ACK>
LIS	<STX>2P 0 MedicalRecordSN1 HosNo1zhangsan 19800101^0^Y F Nation1 Address1 B TelNo1 SecuritySN1 PatientType1 Area1 BedNo1 <CR><ETX>C9<CR><LF>
Instrument	<ACK>
LIS	<STX>3O 1 1^1 BarCode1 TP^TP^1.000000^\\UREA^UREA^1.000000^\\ALT^ALT^1.000000^ R 20170101000000 20171019134827 1.0 20171019134827 SampleType1 RequestDoctor1 RequestDept1 1.000000 Q <CR><ETX>DD<CR><LF>
Instrument	<ACK>
LIS	<STX>4L 1 N<CR><ETB>1B<CR><LF>
Instrument	<ACK>
LIS	<EOT>

3.3.3.2. Sample number query

Instrument Query

Send	Content
Instrument	<ENQ>
LIS	<ACK>
Instrument	<STX>1H ^& ^^ RQ 1394-97 20171019154452<CR><ETX>4C<CR><LF>
LIS	<ACK>
Instrument	<STX>2Q 1 1^1 ALL O<CR><ETX>7C<CR><LF>
LIS	<ACK>
Instrument	<STX>3L 1 N<CR><ETB>1A<CR><LF>
LIS	<ACK>
Instrument	<EOT>

LIS system returns the query result

Send	Content
LIS	<ENQ>
Instrument	<ACK>
LIS	<STX>1H ^& ^^ Q 1394-97 20171019154452<CR><ETX>3B<CR><LF>
Instrument	<ACK>
LIS	<STX>2P 0 MedicalRecordSN1 HosNo1 zhangan 19800101^0^Y F Nation1 Address1 B TelNo1 SecuritySN1 PatientType1 Area1 BedNo1 <CR><ETX>C9<CR><LF>
Instrument	<ACK>
LIS	<STX>3O 1 1^1&S&1 BarCode1 TP^TP^1.000000^\\UREA^UREA^1.000000^\\ALT^ALT^1.000000^ R 20170101000000 20171019134827 1.0 20171019134827 SampleType1 RequestDoctor1 RequestDept1 1.000000 Q <CR><ETX>DD<CR><LF>
Instrument	<ACK>
LIS	4L 1 N<CR><ETB>1B<CR><LF>
Instrument	<ACK>
LIS	<EOT>

3.3.4. Return query samples or download samples

If the LIS system returns the query sample information, the value of the 26th field of O record is "Q"; if the LIS system actively sends the sample information to the biochemical analyzer, the value of the 26th field of O record is "O".

Example

Send	Content
LIS	<ENQ>
Instrument	<ACK>
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LIS	<EOT>